

CLMS to Copernicus Data Space Ecosystem Migration

Weekly Status Update



Author: **European Environment Agency (EEA)**

Date: **2026-07-16**

Version: **1.0.0**

Table of contents

1. Changes since last week	2
2. Status Overview	3
3. CLMS Product on CDSE	3
4. Services	7
5. Methodology	7

Contact:

European Environment Agency (EEA)
Kongens Nytorv 6
1050 Copenhagen K
Denmark
<https://land.copernicus.eu/>

```

// — Load data —
data = await FileAttachment("DATA/migration_data.json").json()
groups = data.groups
summary = data.summary
portfolio = data.portfolio
cdse = data.cdse
lastUpdated = data.scraped_at.slice(0,10)

// — Filter groups —
filtered = groups.filter(d => {
  if (!activeComponents.has(d.component)) return false
  if (searchTerm && !d.name.toLowerCase().includes(searchTerm.toLowerCase())) return false
  return true
})

// — Aggregate by category —
categories = [...new Set(filtered.map(d => d.category).filter(Boolean))].sort()

// — Totals reflecting the current filter selection —
filteredTotals = ({
  datasets: d3.sum(filtered, d => d.total),
  groups: filtered.length
})
catStats = categories.map(cat => {
  const g = filtered.filter(d => d.category === cat)
  return {
    name: cat,
    groups: g,
    total: d3.sum(g, d => d.total),
    on_cdse: d3.sum(g, d => d.on_cdse),
    pct: Math.round(d3.sum(g, d => d.on_cdse) / Math.max(d3.sum(g, d => d.total), 1) * 100),
    s3: d3.sum(g, d => d.s3),
    stac: d3.sum(g, d => d.stac),
    openeo: d3.sum(g, d => d.openeo),
    shub: d3.sum(g, d => d.shub),
  }
})

// — Helpers —
pctColor = (pct) => pct ≥ 100 ? "#00aa00" : pct ≥ 75 ? "#66cc00" : pct ≥ 50 ? "#ffcc00" : pct
≥ 25 ? "#ff8800" : "#ff0000"

pctBar = (pct) => {
  const capped = Math.min(pct, 100)
  const color = pctColor(pct)
  const filled = "■".repeat(Math.floor(capped / 20))
  const empty = "░".repeat(Math.max(0, 5 - Math.floor(capped / 20)))
  return htl.html`<span style="color:${color};font-weight:bold">${pct}%</span> <span
style="color:#bbb">${filled}${empty}</span>`
}

badge = (val, total) => {
  if (val === 0 && total === 0) return htl.html`<span style="background:#888;color:#fff;padding:1px
6px;border-radius:3px;font-weight:bold">—</span>`
  const pct = Math.min(val / Math.max(total, 1), 1)

```

```
const color = pct ≥ 1 ? "#00aa00" : pct ≥ 0.75 ? "#66cc00" : pct ≥ 0.5 ? "#ffcc00" : pct ≥ 0.25 ? "#ff8800" : "#ff0000"
return htl.html`<span style="background:${color};color:#fff;padding:1px 6px;border-radius:3px;font-weight:bold">${val}</span>`
}

check = (on) ⇒ on
? htl.html`<span style="color:#00aa00;font-weight:bold">✓</span>`
: htl.html`<span style="color:#ccc">×</span>`

// — Changes since last week —
changes = data.changes_since_last_week || []
newGroups = changes.filter(d ⇒ d.newly_on_cdse)
positiveChanges = changes.filter(d ⇒ d.s3 > 0 && !d.newly_on_cdse)
lastWeek = changes.length > 0 ? "since " + data.scraped_at.slice(0,10) : ""
odataDelta = data.odata_delta || 0

// — Toggle functions for expand/collapse (plain DOM, no OJS reactivity) —
toggleScript = {
  window.toggleCategory = function(catName) {
    const key = catName.replace(/"/g, '&quot;');
    const groupRows = document.querySelectorAll('tr[data-category="' + key + '"]:not([data-group])');
    const isExpanded = groupRows.length > 0 && groupRows[0].style.display !== 'none';
    groupRows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    if (isExpanded) {
      // Collapsing the category also collapses any open dataset rows within it
      document.querySelectorAll('tr[data-category="' + key + '"][data-group]').forEach(function(r) { r.style.display = 'none'; });
      document.querySelectorAll('span[data-group-arrow="' + key + '"]').forEach(function(a) { a.textContent = '▶'; });
    }
    const arrow = document.querySelector('span[data-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
  window.toggleGroup = function(groupKey) {
    const key = groupKey.replace(/"/g, '&quot;');
    const rows = document.querySelectorAll('tr[data-group="' + key + '"]');
    const isExpanded = rows.length > 0 && rows[0].style.display !== 'none';
    rows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    const arrow = document.querySelector('span[data-group-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
}
```

1. Changes since last week

```
html`
${newGroups.length > 0 ? htl.html`
<div style="background:#e8f5e9;border:1px solid #a5d6a7;border-radius:6px;padding:12px;margin:12px 0">
  <strong>🆕 Newly onboarded CLMS products ${lastWeek}</strong> ${newGroups.map(d ⇒ d.name).join(",
```

```
    ")}
  </div>` : ""}
  ${positiveChanges.length > 0 ? htl.html`
  <div style="background:#e3f2fd;border:1px solid #90caf9;border-radius:6px;padding:12px 12px 4px
  12px;margin:12px 0">
    <strong>📄 More data ingested:</strong>
    ${positiveChanges.map(d => htl.html`<div style="margin:4px 0">${d.name}: <strong>+${d.s3}</
    strong> S3, <strong>+${d.stac}</strong> STAC, <strong>+${d.openeo}</strong> OpenEO</div>`)}
  </div>` : ""}
  ${changes.length === 0 ? htl.html`
  <div style="background:#f5f5f5;border:1px solid #ddd;border-radius:6px;padding:12px;margin:12px
  0;color:#888">No new CLMS products were onboarded since last week.</div>` : ""}
  ${odataDelta === 0 ? htl.html`
  <div
    style="background:#fff3e0;border:1px solid #ffcc80;border-
    radius:6px;padding:12px;margin:12px 0">
    <strong>📄 New CLMS files ingested since last week:</strong> ${odataDelta > 0 ? "+" :
    ""}${odataDelta.toLocaleString()}
  </div>` : ""}
  `
```

2. Status Overview

The CLMS portfolio is composed of `{ojs} filteredTotals.datasets` datasets in `{ojs} filteredTotals.groups` product groups (current selection). Of these, the following are available on the CDSE:

Metric	Value
CLMS in S3	<code>{ojs} cdse.total_dataset_ids</code> dataset IDs
CLMS via OData	<code>{ojs} cdse.total_products_odata.toLocaleString()</code> <code>{ojs} odataDelta > 0 ? "(+ " + odataDelta.toLocaleString() + ")" : odataDelta < 0 ? "(" + odataDelta.toLocaleString() + ")" : ""</code> files
CLMS via STAC	<code>{ojs} cdse.total_stac_collections</code> collections
CLMS via OpenEO	<code>{ojs} cdse.total_openeo_collections</code> collections

3. CLMS Product on CDSE

```
// Toggle-button style filter: click a component to enable/disable it
// (multi-select, all enabled by default). Replaces the old single-select
// dropdown, which only let one component be viewed at a time.
viewof activeComponents = {
  const allComponents = [...new Set(groups.map(d => d.component).filter(Boolean))].sort()
  const active = new Set(allComponents)

  const container = document.createElement("div")
  container.style.cssText = "display:flex;gap:8px;flex-wrap:wrap;align-items:center;margin-
  bottom:12px";

  function render() {
```

```
container.innerHTML = "";
allComponents.forEach(comp => {
  const isActive = active.has(comp);
  const btn = document.createElement("button");
  btn.type = "button";
  btn.textContent = comp;
  btn.style.cssText = `padding:6px 14px;border-radius:16px;cursor:pointer;font-size:0.85em;border:1px solid ${isActive ? "#0d6efd" : "#ccc"};background:${isActive ? "#0d6efd" : "#f5f5f5"};color:${isActive ? "#fff" : "#555"}`;
  btn.addEventListener("click", () => {
    if (active.has(comp)) active.delete(comp);
    else active.add(comp);
    render();
    container.value = new Set(active);
    container.dispatchEvent(new CustomEvent("input"));
  });
  container.appendChild(btn);
});
render();
container.value = new Set(active);
return container;
}
```

```
viewof searchTerm = Inputs.text({label: "Search", placeholder: "Product group...", width: 250})
```

```
// — Build the table via plain DOM APIs —
// so that column widths defined on <colgroup> are guaranteed to apply to
// every row, including rows toggled between hidden/visible.
productGroupsTable = {
  const COLS = 8;
  const colWidths = ["auto", "9%", "9%", "9%", "9%", "9%", "9%", "18%"];

  function makeCell(tag, html, opts = {}) {
    const el = document.createElement(tag);
    if (html instanceof Node) el.appendChild(html);
    else el.innerHTML = html;
    el.style.padding = opts.padding ?? "6px";
    el.style.whiteSpace = "nowrap";
    el.style.overflow = "hidden";
    el.style.textOverflow = "ellipsis";
    if (opts.center) el.style.textAlign = "center";
    if (opts.bold) el.style.fontWeight = "bold";
    if (opts.title) el.title = opts.title;
    return el;
  }

  const wrap = document.createElement("div");

  const table = document.createElement("table");
  table.style.cssText = "width:100%;border-collapse:collapse;font-size:0.8em;table-layout:fixed";

  const colgroup = document.createElement("colgroup");
  colWidths.forEach(w => {
```

```
const col = document.createElement("col");
col.style.width = w;
colgroup.appendChild(col);
});
table.appendChild(colgroup);

const thead = document.createElement("thead");
const headRow = document.createElement("tr");
headRow.style.cssText = "background:#f0f0f0;text-align:left";
["Category", "Total", "CDSE", "S3", "STAC", "OpenEO", "S.Hub", "Progress"].forEach((label, i)
⇒ {
  const th = makeCell("th", label, { center: i > 0 && i < 7, padding: "6px" });
  th.style.borderBottom = "2px solid #ddd";
  headRow.appendChild(th);
});
thead.appendChild(headRow);
table.appendChild(thead);

const tbody = document.createElement("tbody");

catStats.forEach(cat ⇒ {
  const tr = document.createElement("tr");
  tr.style.cssText = "background:#f5f5f5;border-bottom:1px solid #ccc;cursor:pointer";

  const nameCell = makeCell("td", "", { bold: true, title: cat.name });
  const arrow = document.createElement("span");
  arrow.dataset.arrow = cat.name;
  arrow.textContent = "►";
  nameCell.appendChild(arrow);
  nameCell.append(" " + cat.name);
  tr.appendChild(nameCell);

  tr.appendChild(makeCell("td", String(cat.total), { center: true, bold: true }));
  tr.appendChild(makeCell("td", String(cat.on_cdse), { center: true, bold: true }));
  tr.appendChild(makeCell("td", badge(cat.s3, cat.total), { center: true }));
  tr.appendChild(makeCell("td", badge(cat.stac, cat.total), { center: true }));
  tr.appendChild(makeCell("td", badge(cat.openeo, cat.total), { center: true }));
  tr.appendChild(makeCell("td", badge(cat.shub, cat.total), { center: true }));
  tr.appendChild(makeCell("td", pctBar(cat.pct), {}));

  tr.addEventListener("click", () ⇒ window.toggleCategory(cat.name));
  tbody.appendChild(tr);

  cat.groups.forEach(g ⇒ {
    const groupKey = cat.name + "::" + g.name;
    const gtr = document.createElement("tr");
    gtr.dataset.category = cat.name;
    gtr.style.cssText = "border-bottom:1px solid #ddd;display:none;cursor:pointer";

    const gNameCell = makeCell("td", "", { title: g.name, padding: "4px 6px 4px 24px" });
    const hasDatasets = g.datasets && g.datasets.length > 0;
    if (hasDatasets) {
      const gArrow = document.createElement("span");
      gArrow.dataset.groupArrow = groupKey;
      gArrow.textContent = "►";
```

```
gArrow.style.marginRight = "4px";
gNameCell.appendChild(gArrow);
}
const link = document.createElement("a");
link.href = g.url;
link.target = "_blank";
link.textContent = g.name;
link.addEventListener("click", (e) => e.stopPropagation());
gNameCell.appendChild(link);
gtr.appendChild(gNameCell);

gtr.appendChild(makeCell("td", String(g.total), { center: true, bold: true, padding: "4px
6px" }));
gtr.appendChild(makeCell("td", String(g.on_cdse), { center: true, bold: true, padding:
"4px 6px" }));
gtr.appendChild(makeCell("td", badge(g.s3, g.total), { center: true, padding: "4px 6px" }));
gtr.appendChild(makeCell("td", badge(g.stac, g.total), { center: true, padding: "4px 6px" }));
gtr.appendChild(makeCell("td", badge(g.openeo, g.total), { center: true, padding: "4px
6px" }));
gtr.appendChild(makeCell("td", badge(g.shub, g.total), { center: true, padding: "4px 6px" }));
gtr.appendChild(makeCell("td", pctBar(g.pct), { padding: "4px 6px" }));

if (hasDatasets) {
  gtr.addEventListener("click", () => window.toggleGroup(groupKey));
}
tbody.appendChild(gtr);

(g.datasets || []).forEach(ds => {
  const dtr = document.createElement("tr");
  dtr.dataset.category = cat.name;
  dtr.dataset.group = groupKey;
  dtr.style.cssText = "border-bottom:1px solid #eee;display:none;background:#fafafa";

  dtr.appendChild(makeCell("td", ds.id, { padding: "3px 6px 3px 44px", title: ds.id }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.s3), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.stac), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.openeo), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.shub), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));

  tbody.appendChild(dtr);
});
});

// — Summary row (col sums over the currently filtered groups) —
const sumRow = document.createElement("tr");
sumRow.style.cssText = "background:#e0e0e0;border-top:2px solid #999;font-weight:bold";
const sumName = makeCell("td", "Total", { bold: true });
sumName.style.fontStyle = "italic";
sumRow.appendChild(sumName);

const sumTotal = d3.sum(catStats, d => d.total);
```



```
const sumCDSE = d3.sum(catStats, d => d.on_cdse);
const sumS3 = d3.sum(catStats, d => d.s3);
const sumSTAC = d3.sum(catStats, d => d.stac);
const sumOEO = d3.sum(catStats, d => d.openeo);
const sumSHub = d3.sum(catStats, d => d.shub);
const sumPct = Math.round(sumCDSE / Math.max(sumTotal, 1) * 100);

sumRow.appendChild(makeCell("td", String(sumTotal), { center: true, bold: true }));
sumRow.appendChild(makeCell("td", String(sumCDSE), { center: true, bold: true }));
sumRow.appendChild(makeCell("td", badge(sumS3, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", badge(sumSTAC, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", badge(sumOEO, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", badge(sumSHub, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", pctBar(sumPct), {}));
tbody.appendChild(sumRow);

table.appendChild(tbody);
wrap.appendChild(table);
return wrap;
}

html.html`
<div style="margin-bottom:8px;color:#888;font-size:0.9em">
  ${filtered.length} groups in ${catStats.length} categories
</div>
${productGroupsTable}
`
```

4. Services

- [S3 CSV Catalogue](#)
- [OData API](#)
- [STAC Browser](#)
- [OpenEO](#)
- [Sentinel Hub](#)
- [CDSE Documentation](#)

5. Methodology

Per-service counts are matched by:

- **S3:** Direct catalogue count per product type
- **STAC:** Fuzz-matched by checking if CDSE dataset ID (minus `_cog/_nc` suffix) appears in STAC collection IDs
- **OpenEO:** Matched by constructing `CLMS_{id.upper()}` convention
- **Sentinel Hub:** Estimated from S3 availability